GATE ME 90 Days Challenge

Day 45/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
General Aptitude	**Quant II & Data Interpretation**	**59 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on permutations and combinations (repeated letters, restrictions, circular arrangements, selections), basic and conditional probability (addition rule, independent events, Bayes), mensuration of plane figures and solids, and reading tables, bar, line and pie charts. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

The number of distinct arrangements of all the letters of the word BANANA is

(A) 720
(B) 120
(C) 60
(D) 30

Q2 NAT

A committee of 4 is to be chosen from 6 men and 5 women so that it has at least 2 women. The number of ways is _____

Q3 MCQ

A card is drawn at random from a well-shuffled pack of 52 cards. The probability that it is a king or a heart is

(A) 17/52
(B) 4/13
(C) 1/4
(D) 1/13

Q4 NAT

Two fair dice are thrown. The probability that the sum of the numbers is a prime number (round off to three decimal places) is _____

Q5 MCQ

A rectangular garden 60 m × 40 m has a path 2 m wide running all round it on the outside. The area of the path is

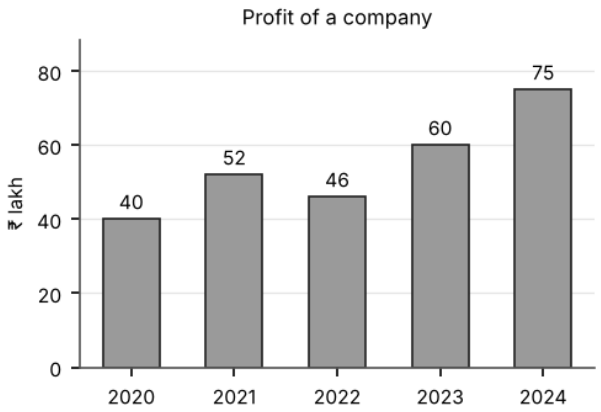
(A) 416 m²
(B) 400 m²
(C) 384 m²
(D) 432 m²

Q6 NAT

A solid metal sphere of radius 3 cm is melted and drawn into a uniform wire of radius 1 mm. The length of the wire (in m) is _____

Q7 MCQ

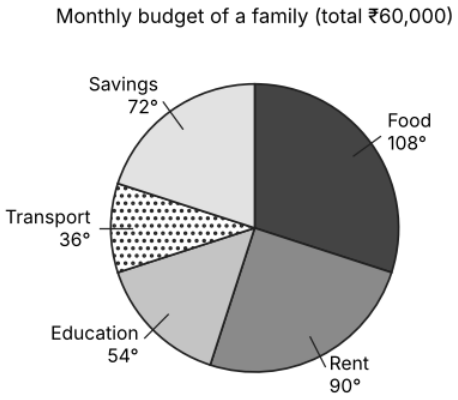
The bar chart shows the profit of a company from 2020 to 2024. The profit in 2024 is about what percentage more than the average profit of the five years?



(A) 25.0 %
(B) 46.7 %
(C) 20.4 %
(D) 37.4 %

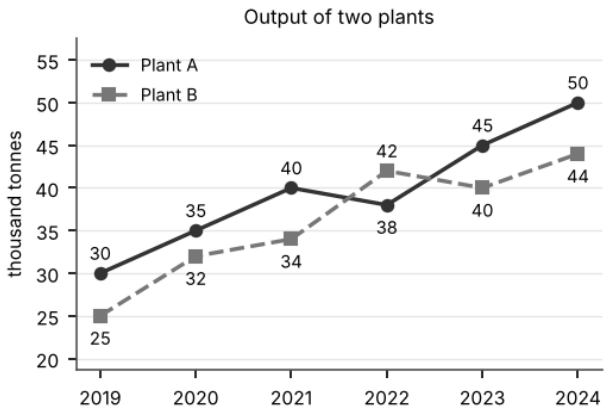
Q8 NAT

The pie chart shows the monthly budget of a family as central angles. By how many rupees per month do the savings exceed the transport spending? _____



Q9 MCQ

The line graph shows the output of two plants A and B. In which year did plant A's output exceed plant B's by the largest percentage (of B's output)?



(A) 2021
(B) 2024
(C) 2019
(D) 2023

Q10 NAT

The table shows the marks of five students. The average total marks (out of 300) of the students who scored more than 75 in Physics is _____

Marks (out of 100) of five students

Student	Maths	Physics	Chemistry
Asha	72	85	64
Bala	58	90	77
Chetan	81	66	70
Divya	95	74	88
Esha	64	79	92

Q11 MCQ

A bag has 3 white and 2 black balls. Two balls are drawn one after the other *with replacement*. The probability that one is white and the other is black is

(A) 6/25
(B) 12/25
(C) 3/10
(D) 3/5

Q12 NAT

The number of ways of distributing 6 different books among 3 students so that each student gets exactly 2 books is _____

Q13 MCQ

The radius of a right circular cylinder is doubled and its height is halved. Its volume

(A) is halved
(B) remains the same
(C) doubles
(D) becomes four times

Q14 NAT

Machine M1 makes 60 % of a factory's output with 2 % defectives, and machine M2 makes the rest with 5 % defectives. An item picked at random is found to be defective. The probability that it was made by M2 (round off to three decimal places) is _____

Q15 MCQ

An equilateral triangle has a perimeter of 36 cm. Its area is about

(A) 62.4 cm²
(B) 72.0 cm²
(C) 54.0 cm²
(D) 124.7 cm²

Answer Key

Q	Type	Answer
Q1	MCQ	(C) 60
Q2	NAT	215
Q3	MCQ	(B) 4/13
Q4	NAT	0.417 (accept 0.414 to 0.420)
Q5	MCQ	(A) 416 m²
Q6	NAT	36
Q7	MCQ	(D) 37.4 %
Q8	NAT	6000
Q9	MCQ	(C) 2019
Q10	NAT	227
Q11	MCQ	(B) 12/25
Q12	NAT	90
Q13	MCQ	(C) doubles
Q14	NAT	0.625 (accept 0.620 to 0.630)
Q15	MCQ	(A) 62.4 cm²

Detailed Solutions

Q1 **Answer: (C) 60**

Concept: n objects with groups of identical ones: n!/(p! q! ...).

BANANA has 6 letters: A three times, N twice, B once.

Arrangements = 6!/(3! × 2!) = 720/12 = **60**

Exam tip: Count the repeats carefully before dividing; a missed repeat doubles or triples the answer.

Q2 **Answer: 215**

Concept: 'At least' problems: add the separate exact cases (or subtract the unwanted cases from the total).

2 women + 2 men: C(5,2) × C(6,2) = 10 × 15 = 150

3 women + 1 man: C(5,3) × C(6,1) = 10 × 6 = 60; 4 women: C(5,4) = 5

Total = 150 + 60 + 5 = **215** (check: C(11,4) − C(6,4) − C(5,1)C(6,3) = 330 − 15 − 100 = 215)

Exam tip: Never fix 2 women first and then choose 'any 2 of the rest': that counts the same committee several times.

Q3 **Answer: (B) 4/13**

Concept: Addition rule: P(A ∪ B) = P(A) + P(B) − P(A ∩ B).

P(king) = 4/52, P(heart) = 13/52, P(king of hearts) = 1/52

P = (4 + 13 − 1)/52 = 16/52 = **4/13**

Exam tip: Option (A) 17/52 is the trap answer that forgets to remove the king of hearts counted twice.

Q4 **Answer: 0.417 (accept 0.414 to 0.420)**

Concept: List the favourable sums and count the outcomes for each.

Prime sums: 2 (1 way), 3 (2), 5 (4), 7 (6), 11 (2): total 15 outcomes

P = 15/36 = 5/12 = **0.417**

Exam tip: Ways to get sum s with two dice = 6 − |s − 7| for s = 2 to 12.

Q5 **Answer: (A) 416 m²**

Concept: Path area = outer rectangle − inner rectangle.

Outer: (60 + 4) × (40 + 4) = 64 × 44 = 2816 m²; inner: 60 × 40 = 2400 m²

Path = 2816 − 2400 = **416 m²**

Exam tip: For a path of width w outside, each dimension grows by 2w; inside, each shrinks by 2w (giving 384 m² here: option C).

Q6 **Answer: 36**

Concept: Volume is conserved when a solid is melted and recast.

Sphere: (4/3)π(3)³ = 36π cm³

Wire: π(0.1)² × L = 0.01πL; so L = 36π/0.01π = 3600 cm = **36 m**

Exam tip: Convert every length to the same unit (here cm) before equating volumes.

Q7 **Answer: (D) 37.4 %**

Concept: Percentage more = (value − base)/base × 100, with the average as the base.

Average = (40 + 52 + 46 + 60 + 75)/5 = 273/5 = 54.6

(75 − 54.6)/54.6 × 100 = **37.4 %**

Exam tip: 25 % is the rise over 2023 only; read which base the question asks for.

Q8 **Answer: 6000**

Concept: Amount of a sector = (angle/360°) × total.

Savings − Transport = (72° − 36°) = 36°

36/360 × 60 000 = **₹6000**

Exam tip: Work with the difference of angles first; one multiplication then gives the answer.

Q9 **Answer: (C) 2019**

Concept: Percentage excess = (A − B)/B × 100; a large absolute gap need not be a large percentage gap.

2019: 5/25 = 20 %; 2020: 3/32 = 9.4 %; 2021: 6/34 = 17.6 %; 2022: B is ahead; 2023: 5/40 = 12.5 %; 2024: 6/44 = 13.6 %

Largest: **2019** (20 %)

Exam tip: The biggest absolute gaps (2021 and 2024, 6 thousand tonnes) are the traps here.

Q10 **Answer: 227**

Concept: Filter the rows first, then average the required quantity.

Physics > 75: Asha (85), Bala (90), Esha (79)

Totals: Asha 72 + 85 + 64 = 221; Bala 58 + 90 + 77 = 225; Esha 64 + 79 + 92 = 235

Average = (221 + 225 + 235)/3 = 681/3 = **227**

Exam tip: Divya scored 74 in Physics, so she is just excluded: read the cut-off ('more than') carefully.

Q11 **Answer: (B) 12/25**

Concept: With replacement, the draws are independent; the two orders (WB and BW) are separate cases.

P(WB) = (3/5)(2/5) = 6/25; P(BW) = (2/5)(3/5) = 6/25

P = 6/25 + 6/25 = **12/25**

Exam tip: Without replacement the answer would be 2 × (3/5)(2/4) = 3/5 (option D).

Q12 **Answer: 90**

Concept: Distributing distinct objects into labelled groups of fixed sizes: $n!/(n_1!\, n_2!\, n_3!)$.

First student: C(6,2) = 15; second: C(4,2) = 6; third: C(2,2) = 1

Ways = 15 × 6 × 1 = 6!/(2! 2! 2!) = **90**

Exam tip: If the groups were unlabelled (just three piles), divide by 3! to get 15.

Q13 **Answer: (C) doubles**

Concept: $V = \pi r^2 h$, so $V \propto r^2 h$.

New volume $\propto (2r)^2 \times (h/2) = 4r^2 \times h/2 = 2r^2 h$

The volume **doubles**.

Exam tip: The curved surface area $2\pi rh$ stays the same in this case: (2r)(h/2) = rh.

Q14 **Answer: 0.625 (accept 0.620 to 0.630)**

Concept: Bayes' theorem: P(M2 | D) = P(M2)P(D | M2)/P(D).

P(D) = 0.6 × 0.02 + 0.4 × 0.05 = 0.012 + 0.020 = 0.032

P(M2 | D) = 0.020/0.032 = **0.625**

Exam tip: A tree with the joint probabilities (0.012 and 0.020) makes Bayes problems one division.

Q15 **Answer: (A) 62.4 cm²**

Concept: Area of an equilateral triangle of side a = (√3/4)a².

a = 36/3 = 12 cm

Area = (√3/4) × 144 = 36√3 = **62.35 cm²**

Exam tip: Its height is (√3/2)a = 10.39 cm; ½ × base × height gives the same area as a check.

What next?

Day 45 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

Dig Career Thrust app

Telegram: daily PDFs

Next, Day 46/90: Heat Transfer, Radiation & Heat Exchangers. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.